

HOW SECURE IS YOUR GSM NETWORK?

GSM services offer protection to the users and network operators against undesirable intrusion by third parties. Let's delve deep into how the security of GSM services is ensured

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The idea of cell-based mobile radio systems originated at Bell Laboratories, USA, in the early 1970s. However, mobile cellular systems were not introduced for commercial use until the 1980s. During the early 1980s, analogue cellular telephone systems experienced a very

rapid growth in Europe, particularly in Scandinavia and the UK, France and Germany. Today, cellular systems are still one of the fastest-growing telecommunication systems.

The global system for mobile communications (GSM) is a digital cellular communications system. In the beginning of cellular systems development, each country had its own system, which was an undesirable situation. In order to overcome this, the Conference of European Posts and Telecommunications (CEPT) in 1982 formed the 'Group Special Mobile' (GSM) in order to develop a pan-European mobile cellular radio system (GSM later became the acronym for 'global system for mobile communications').

In 1989, the responsibility for GSM specifications passed from CEPT to the European Telecommunications Standards Institute. GSM specifications describe the functionality and interface for each component of the system, and provide guidance on the system design. These specifications also standardise the system in order to guarantee proper working between different elements of the GSM system. In 1990, phase-1 of GSM specifications were published but the commercial use of GSM did not start until mid-1991.

At present, GSM operates in 900MHz and 1800MHz bands in Europe, and 850MHz and

1900MHz bands in the US. Together with international roaming and support for a variety of services such as mobile telephony, data transfer, fax, short message service and supplementary services, GSM comes close to fulfilling the requirements for a personal communication system—close enough that it is being used as a basis for the next-generation mobile communications technology.

Architecture of the GSM network

Fig. 1 shows the architecture of the GSM network along with the network interfaces. The GSM network is made up of three sub-systems: mobile station (MS), base station sub-system (BSS) and network and switching sub-system (NSS) comprising a mobile switching centre (MSC) and associated registers. The interfaces defined between each of these sub-systems include U_m air interface between the MS and BSS, A_{bs} interface between the base transceiver station (BTS) and base station controller (within BSS), and 'A' interface between the BSS and NSS.

Mobile station. The MS comprises mobile equipment (ME) and a smart-card called the subscriber identity module (SIM) used by a subscriber to connect to the network. The ME is uniquely identified by International Mobile Equipment Identity (IMEI) number.

The SIM card stores permanent and temporary data about the mobile and the subscriber. The network contains International Mobile Subscriber



Tower for GSM network